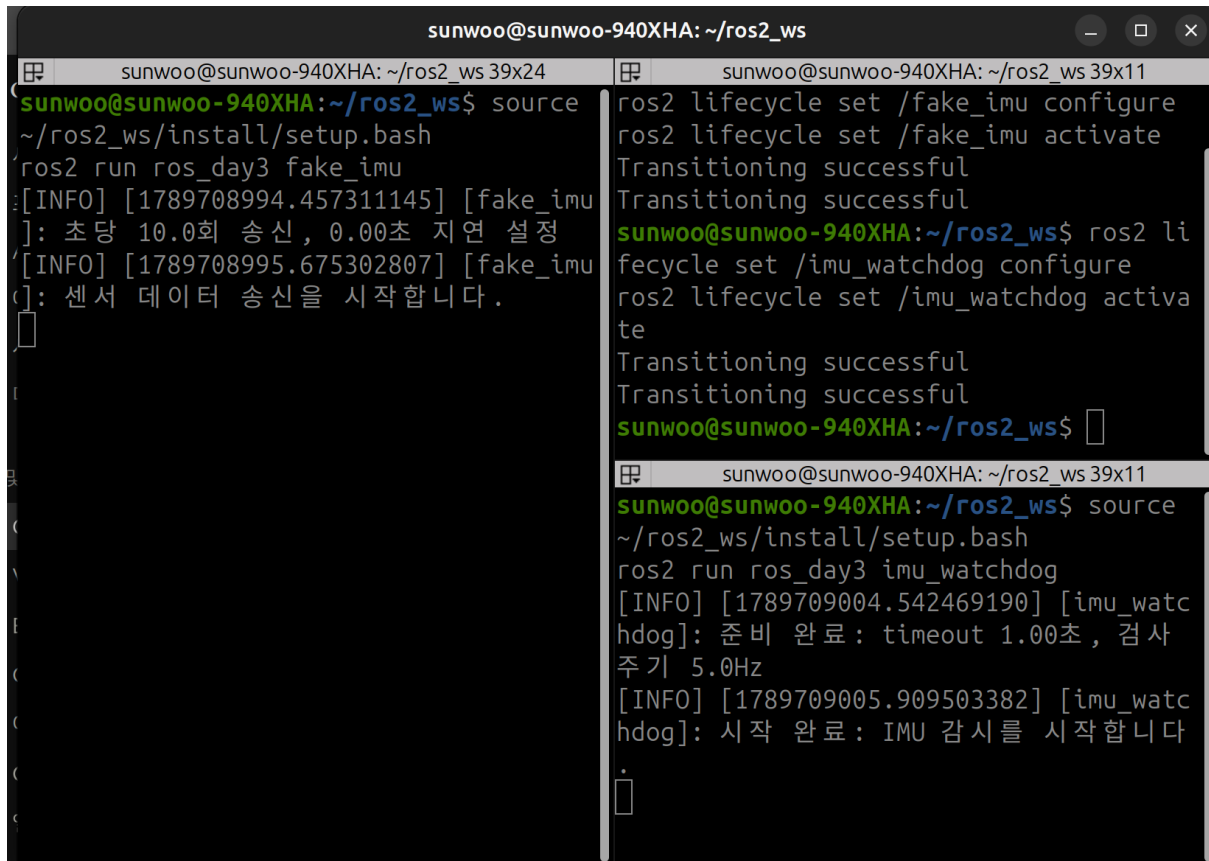


Day4_hw1 증거자료 보고서

김선우



```
sunwoo@sunwoo-940XHA: ~/ros2_ws
sunwoo@sunwoo-940XHA: ~/ros2_ws 39x24
sunwoo@sunwoo-940XHA:~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 fake_imu
[INFO] [1789708994.457311145] [fake_imu]: 초당 10.0회 송신, 0.00초 지연 설정
[INFO] [1789708995.675302807] [fake_imu]: 센서 데이터 송신을 시작합니다.
sunwoo@sunwoo-940XHA:~/ros2_ws$ ros2 lifecycle set /fake_imu configure
ros2 lifecycle set /fake_imu activate
Transitioning successful
Transitioning successful
sunwoo@sunwoo-940XHA:~/ros2_ws$ ros2 lifecycle set /imu_watchdog configure
ros2 lifecycle set /imu_watchdog activate
Transitioning successful
Transitioning successful
sunwoo@sunwoo-940XHA:~/ros2_ws$
sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11
sunwoo@sunwoo-940XHA:~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 imu_watchdog
[INFO] [1789709004.542469190] [imu_watchdog]: 준비 완료: timeout 1.00초, 검사 주기 5.0Hz
[INFO] [1789709005.909503382] [imu_watchdog]: 시작 완료: IMU 감시를 시작합니다
sunwoo@sunwoo-940XHA:~/ros2_ws$
```

fake_imu와 imu_watchdog을 configure→activate함.

activate 직후, 아직 데드락 증상이 로그에 드러나기 전 시점

```
sunwoo@sunwoo-940XHA: ~/ros2_ws
sunwoo@sunwoo-940XHA: ~/ros2_ws 39x24
sunwoo@sunwoo-940XHA:~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 fake_imu
[INFO] [1789708994.457311145] [fake_imu]: 초당 10.0회 송신, 0.00초 지연 설정
[INFO] [1789708995.675302807] [fake_imu]: 센서 데이터 송신을 시작합니다.
sunwoo@sunwoo-940XHA:~/ros2_ws 39x11
ros2 lifecycle set /imu_watchdog configure
ros2 lifecycle set /imu_watchdog activate
Transitioning successful
Transitioning successful
sunwoo@sunwoo-940XHA:~/ros2_ws$ ros2 lifecycle get /fake_imu
active [3]
sunwoo@sunwoo-940XHA:~/ros2_ws$ ros2 lifecycle get /imu_watchdog
sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11
sunwoo@sunwoo-940XHA:~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 imu_watchdog
[INFO] [1789709004.542469190] [imu_watchdog]: 준비 완료: timeout 1.00초, 검사 주기 5.0Hz
[INFO] [1789709005.909503382] [imu_watchdog]: 시작 완료: IMU 감시를 시작합니다
```

ros2 lifecycle get /fake_imu는 active [3]으로 즉시 응답했지만, 바로 이어서 실행한
ros2 lifecycle get /imu_watchdog은 응답 없이 멈춰있음

=>데드락

```
sunwoo@sunwoo-940XHA: ~/ros2_ws
sunwoo@sunwoo-940XHA: ~/ros2_ws$ cd ~/ros2_ws
colcon build --packages-select ros_day3
Starting >>> ros_day3
Finished <<< ros_day3 [7.09s]

Summary: 1 package finished [7.22s]
sunwoo@sunwoo-940XHA: ~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 fake_imu
[INFO] [1789709430.546196977] [fake_imu]: 초당 10.0회 송신, 0.00초 지연 설정
[INFO] [1789709431.704515747] [fake_imu]: 센서 데이터 송신을 시작합니다.

sunwoo@sunwoo-940XHA: ~/ros2_ws$ ros2 lifecycle set /fake_imu configure
ros2 lifecycle set /fake_imu activate
Transitioning successful
Transitioning successful
sunwoo@sunwoo-940XHA: ~/ros2_ws$ ros2 lifecycle set /imu_watchdog configure
ros2 lifecycle set /imu_watchdog activate
Transitioning successful
Transitioning successful
sunwoo@sunwoo-940XHA: ~/ros2_ws$

sunwoo@sunwoo-940XHA: ~/ros2_ws$ ros2 run ros_day3 imu_watchdog
[INFO] [1789709453.120324114] [imu_watchdog]: fake_imu state: active
[INFO] [1789709454.120298556] [imu_watchdog]: fake_imu state: active
[INFO] [1789709455.119995902] [imu_watchdog]: fake_imu state: active
[INFO] [1789709456.120010206] [imu_watchdog]: fake_imu state: active
[INFO] [1789709457.119950308] [imu_watchdog]: fake_imu state: active
```

코드를 비동기 방식으로 수정 후, imu_watchdog 로그에 fake_imu state: active가 1초 간격으로 반복 출력되는 것이 확인.


```
sunwoo@sunwoo-940XHA: ~/ros2_ws
Summary: 1 package finished [7.22s]
sunwoo@sunwoo-940XHA:~/ros2_ws$ source ~/ros2_ws/install/setup.bash
ros2 run ros_day3 fake_imu
[INFO] [1789709430.546196977] [fake_imu]: 초당 10.0회 송신, 0.00초 지연 설정
[INFO] [1789709431.704515747] [fake_imu]: 센서 데이터 송신을 시작합니다.
[INFO] [1789709655.622443152] [fake_imu]: 센서 데이터 송신을 멈춥니다.
z: 0.0
linear_acceleration_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
[WARN] [1789709669.719239457] [imu_watchdog]: [STALE] 마지막 데이터로부터 14.1초 경과 (허용: 1.00초)
[WARN] [1789709669.919318802] [imu_watchdog]: [STALE] 마지막 데이터로부터 14.3초 경과 (허용: 1.00초)
[WARN] [1789709670.119580666] [imu_watchdog]: [STALE] 마지막 데이터로부터 14.5초 경과 (허용: 1.00초)
[INFO] [1789709670.120533496] [imu_watchdog]: fake_imu state: inactive
[WARN] [1789709670.319245772] [imu_watchdog]: [STALE] 마지막 데이터로부터 14.7초 경과 (허용: 1.00초)
```

ros2 lifecycle set /fake_imu deactivate 실행 직후, imu_watchdog 로그에 [STALE] 마지막 데이터로부터 14.xx초 경과 경고가 연속 출력되고, 곧이어 fake_imu state: inactive로 상태가 갱신됨.